

1. Moles of both
2. Figure out Limiting Part.

LESSON-6 Thermodynamics

i. Intensive Prop:

- independent mass & size. Molar, Specific.

Extensive Prop:

- Mass Mole Volume

(U) (H) (S) (G) (C)

ii. State works : $\Delta U, q, w \dots$

Path function: heat, work
(q) (w)

$$\Delta U = q + w$$

↓
Internal energy

Internal Energy

Enthalpy

Entropy

Gibbs free Energy

Heat Capacity

$\Delta S = +ve$

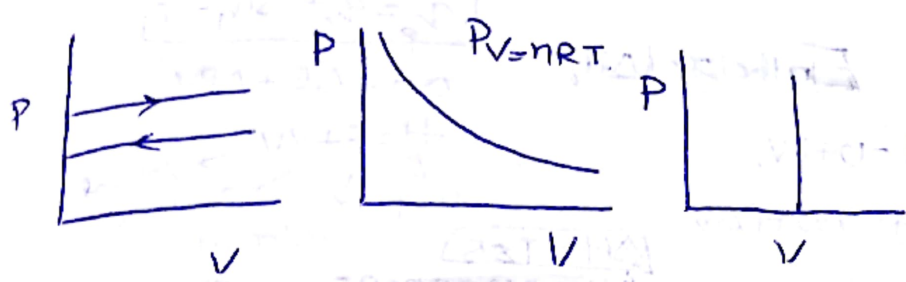
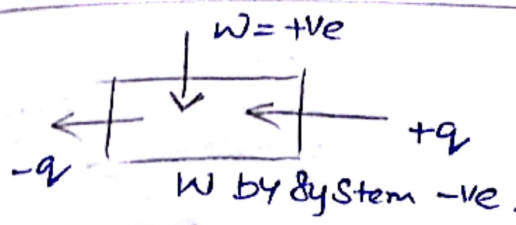
happening suddenly

iii. Spontaneous: done

Non Spontaneous: can't

Occur of its own in
a Particular direction.

$\Delta S = -ve$



ISOCHORIC: $\Delta V = 0$ $V = \text{constant}$

$$\Delta U = q + (P \Delta V)$$

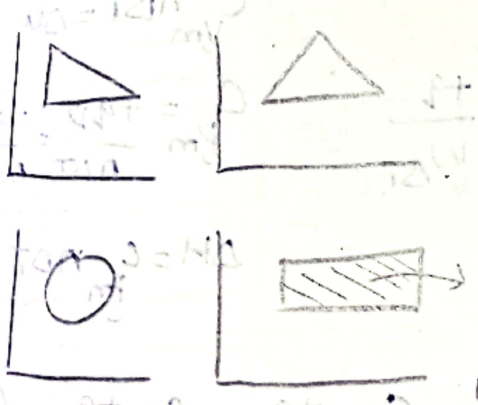
$$\boxed{\Delta U = q_v} \quad \boxed{\Delta H = q_p}$$

Isobaric: $\boxed{W = -P_{\text{ext}} \Delta V}$ $P \rightarrow \text{const}$

Isothermal: $\Delta T = 0$
 $\Delta U = 0$ $\boxed{q = -W}$

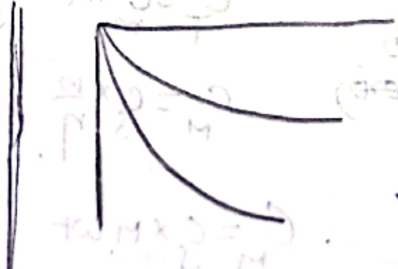
Cyclic Process:
 $\Delta U = 0$ (State)
 $q = -W$

Adiabatic: ($q \rightarrow \text{constant}$)
 $\Delta U = 0 + W_{\text{ad.}}$

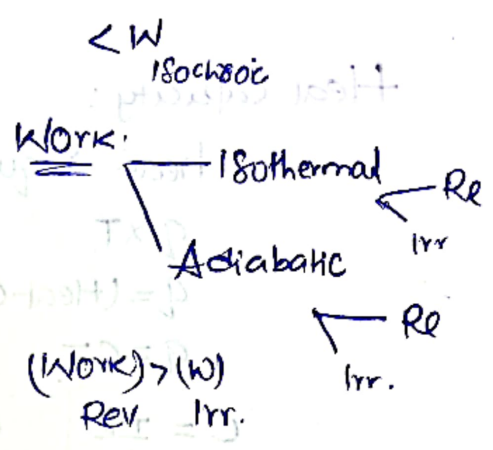


Area is WORK Done

- Loop Clock: $W = -ve$
- Anti clock: $W = +ve$



$W_{\text{Irr}} < W_{\text{Rev}} < W_{\text{Adiabatic}}$
 Isobaric Isothermal Adiabatic



$$W = -P_{\text{ext}} \Delta V$$

Irreversible

$$W_{\text{irr}} = -P_{\text{ext}} \left[\frac{nRT}{P_2} - \frac{nRT}{P_1} \right]$$

$$W_{\text{Reversible}} = -2.303 nRT \log \left(\frac{V_2}{V_1} \right)$$

$$W_{\text{Reversible}} = -2.303 nRT \log \left(\frac{P_1}{P_2} \right)$$

$$\boxed{P_1 V_1 = nRT}$$

$$W_{\text{Re}} = -2.303 P_1 V_1 \log \left(\frac{V_2}{V_1} \right)$$

$W_{\text{irr}} = -P_{\text{ext}} \Delta V \rightarrow \text{Constant Pressure..}$
 Irr Process. $\leftarrow$ (+high Pressure)

if $P_{ext} = 1$.

$$W = -\Delta V.$$

$$W = -\Delta n_g RT.$$

Enthalpy (ΔH)

$$\Delta H = U + PV.$$

$$\Delta H = \Delta U + P\Delta V.$$

(Extensive)

(Enthalpy) (Enthalpy)

ΔH
Constant P
Enthalpy at
Constant V
 ΔE

$$q_p = q_v + \Delta n_g RT$$

$$\Delta H = \Delta E + \Delta PV$$

$$H = E + PV$$

Heat. Internal Energy
Volume Pressure

$$\Delta H = T\Delta S$$

$$\Delta H = \Delta U + \Delta n_g RT.$$

$$P\Delta V = \Delta n_g RT.$$

$$\Delta H > 0, \Delta H < 0$$

$$\Delta n_g > 0, \Delta n_g < 0.$$

Heat Capacity:

$$\Delta H = \Delta U + n_g RT.$$

heat required to rise 1 temp.

$$q = \Delta T.$$

$$q = (\text{Heat Capacity}) T.$$

$$q = CT.$$

$$C = \frac{+q}{\Delta T}$$

$$C_T = \frac{+q}{\Delta T}$$

$$C_M = \frac{+q}{n\Delta T}$$

(Molar Heat)

$$C_{V,M} = \frac{+q_V}{n\Delta T} = \frac{\Delta U}{n\Delta T}$$

$$C_{V,m} n\Delta T = \Delta U.$$

$$C_{P,m} = \frac{+q_P}{n\Delta T} = \frac{\Delta H}{n\Delta T}$$

$$\Delta H = C_{P,m} n\Delta T.$$

$$C_S = \frac{+q}{W\Delta T}$$

(Specific)

$$(i) C_P - C_V = R.$$

$$(ii) \frac{C_P}{C_V} = \gamma \text{ (Poisson's ratio)}$$

$$(iii) C_P - C_V = nR.$$

Isothermal: Heat Capacity (Zero) (Infinite).

$$C_T = \frac{+q}{n\Delta T}, C_M = \frac{+q}{n\Delta T}, C_S = \frac{+q}{W\Delta T}$$

$$C_T = C_M \times n = C_S \times W.$$

$$C_M = C_S \times \frac{W}{n}$$

$$C_M = C_S \times \text{M.wt.}$$

$$\text{Avg } C_{vm} \text{ mx} = \frac{n_1 C_{vm1} + n_2 C_{vm2}}{n_1 + n_2}$$

$$C = \frac{dq}{dT}$$

Adiabatic (Isolated)

$$q=0$$

$$\Delta U = 0 + W$$

$$\Delta U = W_{\text{adiabatic}}$$

$$nC_V \Delta T = W_{\text{ad.}}$$

$$T_V^{\gamma-1} = \text{Constant}$$

$$P_T^{\gamma} = \text{Constant}$$

$$P_V^{\gamma} = \text{Constant}$$

$$T \propto V^{-\frac{1}{\gamma-1}} \text{ find } \gamma?$$

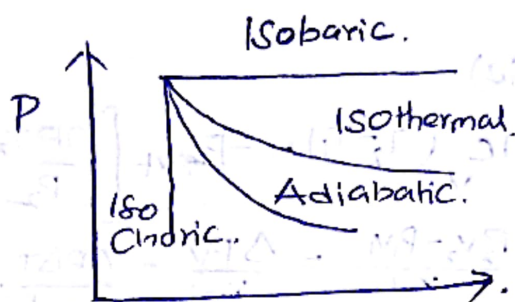
$$T = KV^{-\frac{1}{\gamma-1}}$$

$$T_V^{\gamma-1} = \text{Constant}$$

$$T_V^{\gamma-1} = \dots$$

$$\gamma-1 = \frac{1}{2}$$

$$\gamma = \frac{3}{2}$$



$$\underline{VIP} \odot$$

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -RT \ln K_{eq}$$

$$\Delta U = q + w$$

$$\Delta H = \Delta U + P\Delta V$$

$$\Delta H = nC_P \Delta T$$

Mono	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$
Di	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$
Tri Linear	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$
Tri non Linear	6	$3R$	$4R$	$\frac{4}{3}$
Linear	Degrees of freedom	C_V	C_P	γ

Total Degrees of Freedom : $3N$ (Translation Rotational Vibrational.)

Mono : $3 \times 1 = 3 \rightarrow 3T$

Di : $3 \times 2 = 6 \rightarrow 3T + 2R + 1V$

Tri : $3 \times 3 = 9 \rightarrow 3T + 2R + 4V$

Tri Non Linear : $3T + 3R + 3V = \text{Total } 9 \text{ Rotations } \dots$

Work Done Adiabatic Reversible:

$$W_{irr} = -P_{ext} \Delta T$$

$$\Delta U = nC_V \Delta T = -P_{ext} \Delta V = -P_{ext} \left[\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right]$$

(Adiabatic)

$$n C_V (T_2 - T_1) = -P_{ext} \left[\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right]$$

$$P_{rev} \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \frac{\Delta PV}{\gamma - 1} = \frac{nR \Delta T}{\gamma - 1}$$

$$1 \text{ cal} = 4.2 \text{ J}$$

$$1 \text{ atm} = 101.3 \text{ kPa} = 24.3 \text{ cal} = 101.3 \text{ kPa}$$

→ 100 x Joules

1. Polytropic WD $P V^n = \text{constant } n \neq 1$

$$W = \frac{\Delta PV}{n-1} = \frac{nR \Delta T}{n-1}$$

2. Free Expansions $W = 0$

$$\Delta U, \Delta Q, W \rightarrow \text{Zero}$$

Solid < Liquid < Gas

Entropy (S)



$$\Delta S \propto T_{\text{system}}$$

Degree of Randomness

$$\Delta S \propto \frac{1}{P}$$

$$\Delta S = \frac{+q_{rev}}{T}$$

Disorderliness

$$\Delta S = \frac{q_{rev}}{T} \propto \text{Randomness}$$

Units: JK^{-1} or calK^{-1}

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

$$\Delta S = \sum S_{\text{product}} - \sum S_{\text{reactant}}$$

$$\Delta S = nC_V \ln \frac{V_2}{V_1} + nR \ln \frac{V_2}{V_1} \quad (1)$$

$$\Delta S = nC_P \ln \frac{P_2}{P_1} + nR \ln \frac{P_1}{P_2} \quad (2)$$

$$\Delta S_{\text{System}} (\text{irr Adiabatic}) = 0$$

$$\Delta S = \frac{\Delta H_{\text{fusion, Sublimation}}}{T}$$

$$\Delta G = \Delta H - T \Delta S$$

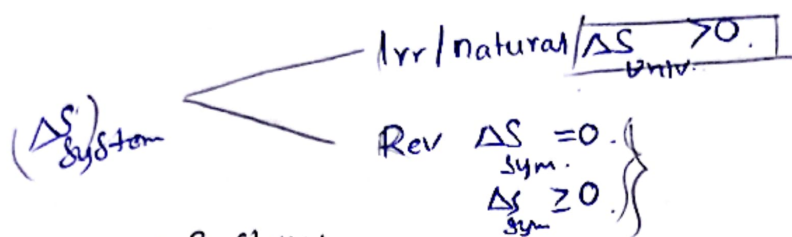
$$\Delta G_{\text{System}} = -T \Delta S_{\text{Univ}}$$

$\Delta G = -ve$ Spontaneous
 $\Delta G = +ve$ non Spontaneous

$$\Delta G = \Delta G^\circ + RT \ln Q$$

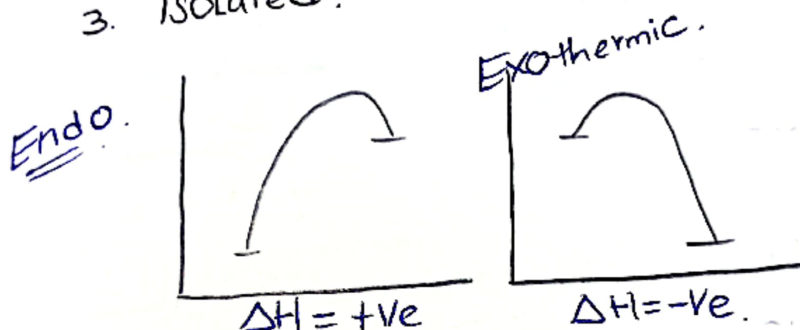
$$\Delta G^\circ = -RT \ln K_{eq}$$

3rd Law: At 0K All rotations can be stopped ...
(-273K)



Types of System:

1. Open.
2. Closed
3. Isolated.



Endothermic: Absorbs heat.

Exothermic: Releases heat.

$$\boxed{\frac{H - H^\circ}{T} = C_p \Delta T} \quad \Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = -RT \ln K = -2.303 RT \log K \dots$$

$$\Delta H = C_p \Delta T$$

$$\Delta U = C_v \Delta T$$

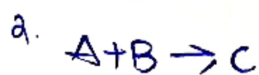
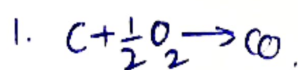
Enthalpy of formation ΔH_f°

Product...
 1 MOLE
 Reactant: Standard State...

↓ State...

C_{graphite}, H₂, O₂, Na₂

ΔH_f°



$$\boxed{\Delta H_f^\circ = \sum a_i \Delta H_f^\circ (\text{Product}) - \sum b_j \Delta H_f^\circ (\text{Reactants})}$$